

The Evidence File

Every result, dataset, correction and negative finding from the work on problem statement 4 — compressed into one reference for building the deck and the video. The pipeline design lives in its own document; this is the evidence behind it.

HOW TO READ THIS

MEASURED

 means a number produced by committed code against ground truth.

FAILED

 means we built it, scored it, and it does not work — reported because a jury values a tested negative over an unchecked claim.

UNVALIDATED

 means built but not yet scoreable, with the reason stated.

1 · The thesis

A modern plant is not short of instruments. The problem is that **the signals plant teams trust are each misleading in a specific, measurable way** — and we can prove each one.

WHAT THE PLANT TRUSTS	WHAT IT REPORTS	WHAT IS ACTUALLY TRUE	EVIDENCE
Utilisation ranking	busiest = bottleneck	busy ≠ constraining	46% vs 35%
The torque reading	joint is in spec	the sensor can be what drifted	11,631 escapes
A tool alarm	that tool is failing	may be the fastener lot	21 of 23
"Speed up the slow station"	recovers output	recovers nothing off-constraint	106 vs 106
The downtime log	breakdowns are the loss	micro-stops dominate and are unlogged	~97/shift

2 · What we built

Simulators

- Line simulator** — 20-station spine + subassembly merge, buffers, breakdowns, three product variants, rework loop, shift calendar with breaks, changeovers, andon stops, PM windows, parallel stations, conveyor transit, micro-stops. Built on **common random numbers**, so a counterfactual re-run differs only by the perturbation.
- Tool simulator** — four tool families (nutrunner, spot-weld, adhesive, press-fit), gamma-process degradation, confounded channels, twelve fault conditions.

Datasets

DATASET	SIZE	CONTENTS
v3/process	190 MB	40 tools × 20,000 ops, 12 fault conditions, held-out healthy calibration set, Wiener-process firewall set, maintenance log, material lots, EOL events
v5/fLow	235 MB	162 runs (156 shifts + 6 three-shift sequences), 4 layouts, 972 tools on the line , dark stations, dropouts, measurement artifacts
v5/truth	1.5 MB	1,248 labelled blocks: sensitivity ground truth, marginal-value vectors, persistence, forecast labels, intervention outcomes

THE UNIFICATION THAT MADE THE HEADLINE CLAIM TESTABLE

Until v5, tool telemetry and line events were separate universes — different VINs, different clocks. That made the **joint stop-or-continue decision** literally untestable. In v5 each run's tools are driven by that run's own vehicle sequence, so (run, station, vin) joins exactly (verified 100% VIN match). Now answerable: "tool at S07 began drifting at 10:14 — was S07 the constraint then?"

Realism the dataset carries

Per-controller **clock skew ± 3 s**, 0.2% dropped scans, 0.05% duplicates, late arrivals, NULL bursts in 35% of tool channels, temporary sensor dropouts, and micro-stops that inflate cycle time with **no state change** — verified invisible in the downtime log (2,648 micro-stops vs 876 logged down entries across 30 runs, zero overlap).

3 · Defect side — the strongest results

Answers the brief's "predict likely defects before they happen" directly.

11,631

genuinely defective joints the controller stamped **OK** under a sensor-bias fault, out of 11,826 truly defective — a **98% escape rate**. The torque reading moved -0.55σ while true clamp force collapsed -3.74σ . Monitoring the spec channel alone is structurally blind to this.

RESULT	FIGURE	
Drift warning before sustained scrap, held-out calibration (per-condition medians; bearing/spread-only faults are the exception — median -292, detected after scrap begins)	421-1,322 vehicles	MEASURED
Sensor-bias escapes (11,826 true defects, 195 flagged)	11,631 / 98%	MEASURED
Pure transducer drift: good parts falsely rejected (1 true defect, 2,356 NOK)	2,355	MEASURED
Bad-lot simultaneity: tools sharing the lot that alarmed together / tools on other lots	21 of 23 / 0	MEASURED
Four failure modes separated by primary/secondary shift ratio	0.87 / 0.17 / 0.01	MEASURED
Remaining-life estimates biased long → coarse bands only, never a countdown	~3×	MEASURED
Sudden step failures — no warning possible from any method	physics	MEASURED

THE FOUR FAILURE MODES, AND WHY IT MATTERS

An alarm is not a work order. **Mechanical wear** (all channels move together) → service the tool. **Sensor bias** (reading barely moves, everything else does) → urgent, it is passing bad joints. **Pure transducer drift** (reading collapses, secondaries flat within $\pm 0.11\sigma$) → **recalibrate only, the tool is fine**. **Bearing wear** (mean unchanged, spread $\times 2.6$) → replace, calibration is useless. Two of these have **opposite economics** and look identical if you watch only torque.

4 · Flow side — mixed, and honestly so

Answers "see where bottlenecks are forming" — the clause the brief leads with, and our weaker half.

Detection MEASURED

METHOD	TOP-1	TOP-2	REGRET (CARS/BLOCK)
Utilisation ranking (<i>what the reference material proposes</i>)	35.1%	48.0%	2.60
Active period (Roser 2001)	46.0%	57.9%	2.24
Effective cycle time (ours)	43.1%	59.4%	2.24

On strong constraints ($\text{margin} \geq 2$ cars, $n=202$). Perfect-picker ceiling 2.27 cars/block, so we capture ~45% of available gain.

Two regimes, reported separately and never averaged. On faulted runs all methods do better (26.9% top-1); on a balanced healthy line ours fall to 13–15% and utilisation to 8.6% — which is *correct*, because a balanced line genuinely has no single dominant bottleneck. A detector reporting "nothing dominates" there is right.

Line physics MEASURED

106 vs 106

We jammed a five-station line, repaired it, then made the slow station **twice as fast**. Output was identical. The queue simply relocated downstream within five minutes, and the station then sat blocked 30% of the time. *An hour saved at a non-bottleneck is a mirage* — Goldratt, 1984.

- **A backlog never clears while input continues** — the buffer stayed at capacity for the whole run and emptied only when we stopped admitting cars, then drained fully in 12 minutes. Breaks and shift ends are the *only* WIP-drain mechanisms.
- **Release-rate control (CONWIP)** — capping cars-in-line gave the **same throughput with 36% lower lead time** and zero equipment change.
- **The constraint moves ~6 times per shift** (up to 25) — including in runs with *no injected fault*, purely from buffer dynamics.
- **Little's Law confirmed** — 8 extra cars of WIP produced exactly 8 minutes of extra lead time at unchanged output.
- **Cost of waiting for the next break: 9 cars**, measured under paired replications — this makes stop-or-continue quantitative.

Forming — one works, one failed

MECHANISM	RESULT	
Buffer countdown	60% of warnings followed by a real block. Median error +0.6 min (unbiased) but IQR –11.9 to +10.6 min — a ranked ordering, not a countdown	WORKS, WITH LIMITS
Drift-based overtake risk	At 70–100% stated confidence the station becomes the constraint 5.9% of the time. Badly overconfident — the second time linear extrapolation of an early trend failed here	FAILED

5 · Dark stations

The brief's hardest think-about prompt, treated as the centrepiece.

88.8s vs 58.0s

Bracketing bias, **measured not asserted**. Scan-to-scan occupancy at a dark station reads 88.8 s while true work time is 58.0 s — a **53% overstatement**, because occupancy includes waiting caused by neighbours. This is exactly why bracketing must be paired with a state estimator rather than used alone.

- 3 permanently dark stations per run; **98 of 162 runs contain consecutive dark pairs**
- Tier ladder A→D: directly observed → strongly estimated → segment-localised → bounded + costed sensor recommendation
- **Localisation built but unvalidatable** UNVALIDATED — only **5 of 142** dark blocks contain the constraint, so a detector always answering "elsewhere" scores 96% from base rate alone. Dark stations are drawn at random while the constraint concentrates on genuinely slow stations, so they rarely coincide. Dataset gap, not a method verdict.

6 · Research that grounds the pitch

FINDING	SOURCE
Breakdowns ~18% of production loss; minor stops 34% ; reduced speed 22%	industry OEE loss taxonomy, automated capture
Micro-stops 15–25% of capacity per shift, invisible in downtime logs	industry micro-stop analyses
Manual assembly cycle-time coefficient of variation 0.25–0.6	assembly task-time variability research
Robotic station 5th–95th percentile spread of 0.94 s over 194 cycles	robotic workstation study
Automotive unplanned downtime above \$2M/hour	Senseye/Siemens, True Cost of Downtime
Best published bottleneck look-ahead ~2.5 min; no accuracy-vs-horizon curves published	C&IE 2024 and surrounding literature

Primitives named, never claimed as invented: active period (Roser, Nakano & Tanaka 2001), CUSUM (Page 1954), Theory of Constraints (Goldratt 1984), model/shadow/twin taxonomy (Kritzing 2018), virtual metrology (semiconductor practice).

7 · The self-audit — six times we caught ourselves

No competing team will say anything like this, and Round 3 is a mentored discussion.

1. **A detector scored 95.2%** — because the ground truth used the detector's own formula. An identity, not a measurement.
2. **Three detectors returned byte-identical scores** — two of them were provably the same statistic with opposite sign.
3. **"Zero false alarms across 35,300 operations"** — true only on the data the thresholds were tuned on. Held-out rate is ~1 per 5 tool-weeks.
4. **The simulated line's first station was permanently saturated**, making it the trivial "bottleneck" in 8 of 28 runs — an artefact, not a finding.
5. **557 of 558 buffer "countdowns" were already-full buffers** — the stage looked productive while predicting nothing.
6. **Dark-block localisation was wrong in both directions** — comparing block traversal against station processing time made it fire on 84% of blocks; correcting to matched intervals made it fire on almost none.

All six fixed, and every headline number quoted here is post-fix. **Ground truth is now economic** — the constraint is the station whose speed-up actually produces more cars, measured under identical random draws — a definition no detector can recompute.

8 · Corrections propagated

WAS	NOW
"zero false alarms" in README, script and report	removed everywhere, replaced with the held-out figure and the reason it was wrong
Tier B bracketing "exact"	"strongly estimated" — we inject ± 3 s clock skew ourselves
Stop-or-continue used a hard <code>1{is the constraint}</code>	takes P(constraint) , matching what the detector actually emits
Cost of continuing multiplied a constant P(defect)	integrates a rising hazard , matching the acceleration premise
"Capital cost of the data layer: zero"	"minimal new sensing hardware", with integration, licences, cybersecurity and validation named
" <code>unit_scan</code> alone is enough"	"carries much of... where coverage is reliable" — our own generator injects rework, dropped scans and parallel stations that break it

9 · Built vs designed — the honest inventory

CAPABILITY	STATE
Event ingestion, state reconstruction, effective-CT detection	BUILT & MEASURED
Multi-channel drift detection with held-out calibration	BUILT & MEASURED
Failure-mode signatures, lot simultaneity, bracketing bias	BUILT & MEASURED
Buffer countdown	BUILT & MEASURED
Drift-based overtake risk	BUILT, FAILED
Dark-block localisation	BUILT, UNVALIDATABLE
Failure-mode classifier as a model; genealogy containment; twin-drift metric	designed, not built
Decision surface as software; stop-or-continue; acknowledge-and-rearm; sensor placement	designed, not built

10 · What we will not claim

- A blended bottleneck accuracy number — quote the two regimes or neither
- "Guaranteed" false-alarm rates — conforal guarantees assume exchangeability that production data violates
- Invention of the primitives — Roser, CUSUM, Goldratt, virtual metrology are all named
- A precise remaining-life countdown, or prediction of sudden failures
- Closed-loop control of a live line
- That a bracketed dark-station time is exact, or that lot simultaneity *proves* causation

THE CLOSING SENTENCE

"The detection primitives are established — Roser, CUSUM, virtual metrology. Our contribution is three things: locating constraints inside unsensored blocks, a joint quality-throughput stopping decision that no existing system can make because flow and quality live in separate tools, and a measured lead-time curve where the published record has almost none."

For the deck — the strongest material

SLIDE	LEAD WITH	WHY IT LANDS
1	106 vs 106. The obvious fix recovers nothing. Plus 56 vehicles lost per shift from an 8-second slip, and 11,631 defects passed as OK.	counter-intuitive, measured, and it sets up the unifying line: <i>the plant's instruments point at the wrong thing</i>
2	The closed loop with a human in it, and three named choices: effective processing time over utilisation; multi-channel then classify the mode; bracketing <i>plus</i> state estimation.	answers "is it live?" and "would a plant let this touch the line?" in one figure
3	Fault injection with T0 known. Detection 46%/58% vs 35%/48%. Drift warning 421–1,322 vehicles. Lot control 21 of 23 vs 0.	early warning is measured , not claimed — because we built the plant as well as the twin

Best single visual: the observability map — a station reading "*2 of 10 signals measured, 6 inferred, confidence 0.81*". No competing team will show one, and it answers the brief's hardest prompt in a single image.

Provenance. Every figure here is produced by committed code against ground truth held in `hidden/` that the detectors never read. Industry figures are cited and marked as ranges. Negative results are reported at the same prominence as positive ones — the six self-audit catches are, on their own, a stronger credibility signal than any single accuracy number.